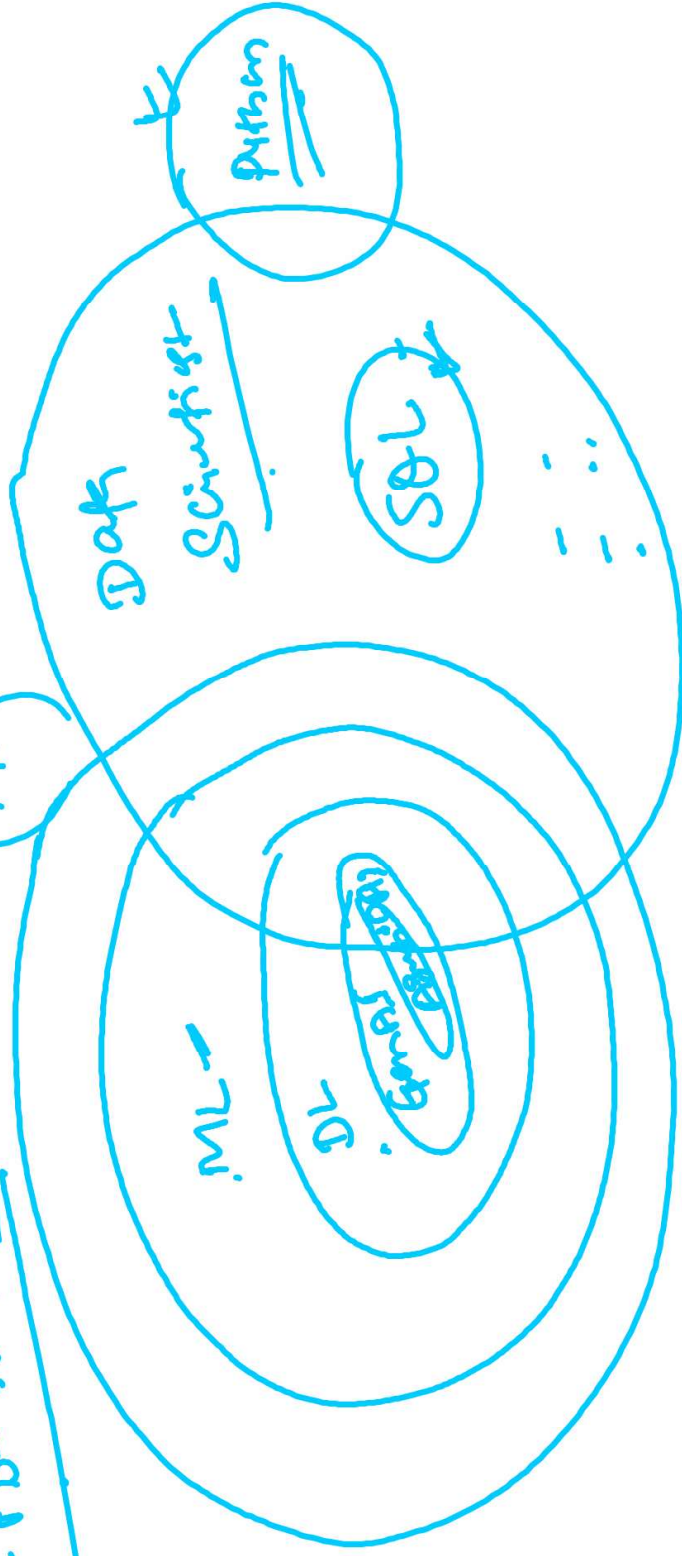


ML - machine learning

Gen AI + Asentis AI

(AI)



$y = \text{yes, when } x_1 + x_2 \geq 2$

x_1	x_2	output
1	1	yes
1	0	no
0	1	no
0	0	no

AI

Rule based AI

ML based AI

very hard to Predict the logic

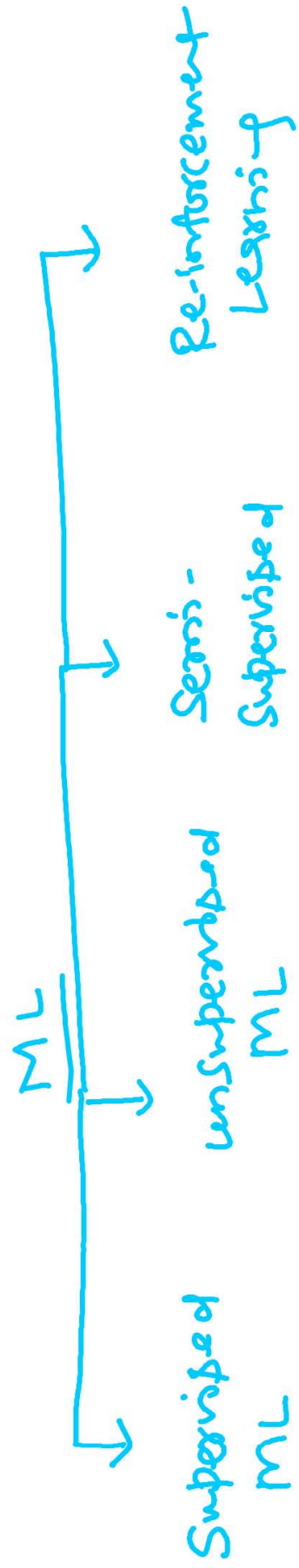
if-else condition

Logic $\rightarrow$ Hard coded rules

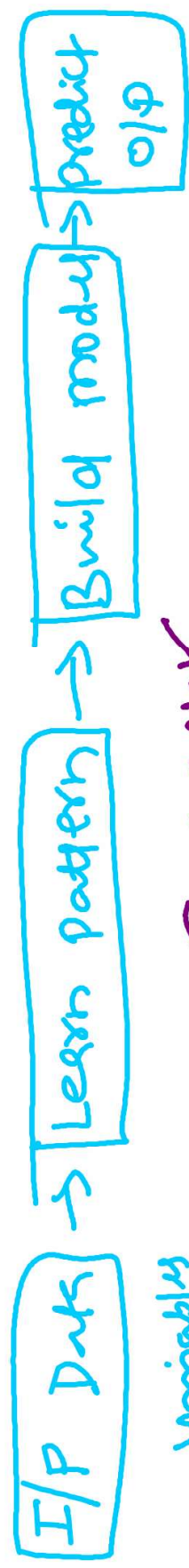
x	y
1	1
2	4
3	9
4	16
5	25
11	121

$y = f(x)$

Square



Labeled data Unlabeled data



Variables

Obs

A	B	O/P	Result ✓
1	2	yes	Output ✓
5	7	NO	Label ✓
0	3	NO	
4	2	yes	

Age	Salary	Buy car
20	6L	NO
28	15L	yes
40	50L	yes
30	60L	yes

Cluster

	Age	Salary	
C1	20	4L	low
✓ C2	22	7L	
✓ C3	25	15L	
	30	40L	high
	28	45L	
C4	45	70L	less
✓ C5	32	90L	less
	50	1.5 cr	less

C2 C3

Target - Pri



- marketing

RM

Group / cluster / segment

Wish

use cases - Supervision

① Fraud detection

Howse

Email - spam/spam

Disease

Supervised ML

Regression

O/p is continuous Number

Delhi
26 HK
 Rohini = 1.5 cr
 Dwarka = 1 cr
 CP = 4.5 cr

Classification

O/p - Category
 - Binary or
 Multiclass

081/yes/no/TF/+-

Movie Rating - 1 2 3 4 5

A B C

Soil/oil

Life cycle

Build Supervised ML

👉 NILOTPAL BANIK + 3 others raised hands

① collect the raw dataset

② Label data - yes ✓

③ Split the data into

Training & Test

basis the labeled dataset

④ Train → model - Algorithm/model build

⑤ Test model

⑥ Predict the new custom dataset

Data Acquisition
data gathering

① Start up
→ website
→ client

② Large/mid-R
own data
→ Big data

③ Client
Share data

ML

Model building

State of the art model

①

Linear Regression $\rightarrow$ Regression

②

polynomial Regression $\rightarrow$ Regression

③

Ridge, Lasso, ElasticNet - Regularization $\xrightarrow{R}$ $\xrightarrow{C}$ model

④

Logistic Regression - Classification

⑤

Decision Tree $\xrightarrow{R}$ $\xrightarrow{C}$

⑥

Bagging - Random Forest $\xrightarrow{R}$ $\xrightarrow{C}$

⑦

Boosting - Ada boost, Gradient boosting, XGBoost, Light GBM, Cat boost $\xrightarrow{R}$ $\xrightarrow{C}$

⑧

Stacking & voting

SOFT
Ensemble
Model

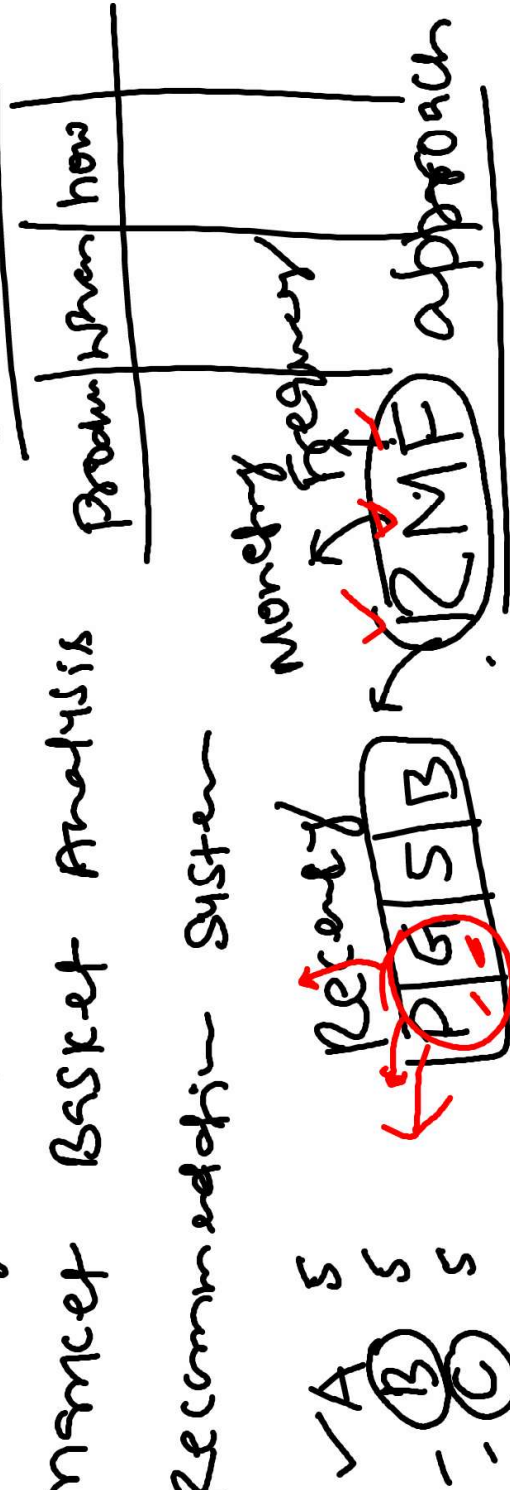
{ K-nearest Neighbors (KNN) - distance $< \frac{R}{C}$
Support vector machine (SVM) $< \frac{R}{C}$
Naive Bayes Theorem - Classification

unsupervised ml

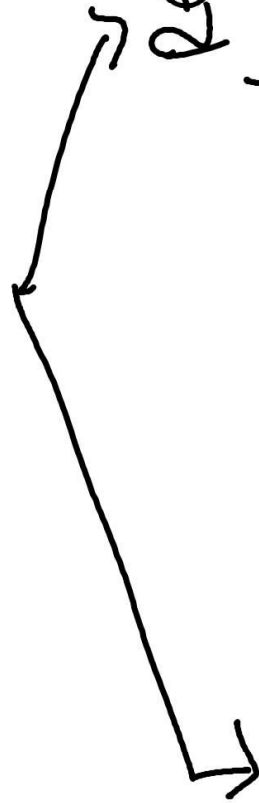
No label data available. Grouping/Clustering

- Customer Segmentation
- Anomaly detection
- Market Basket Analysis
- Recommendation System

Retail - Grocery
Customer



Model - Data Mining topic



Recommendation System

&

Market Basket Analysis

or
Association Rule Mining

Content Based

- Collaborative / User-based Recommendation
- Popularity based Recommendations
- Hybrid based Re -

Clustering
or
Grouping
or
Segmentation